bottom of a valley but is actually more like a divot in the valley wall. It can stay tucked in there for a long time—plenty long enough for the growth of galaxies, the birth of stars, the evolution of life, and the production and distribution of more superhero movies than anyone could ever really want—but the possibility looms that a large enough disturbance could kick it over the edge, and then there would be nothing to stop it from landing on the *actual* valley floor. And that would be really, really, apocalyptically bad. For reasons we will shortly discuss in gory detail.

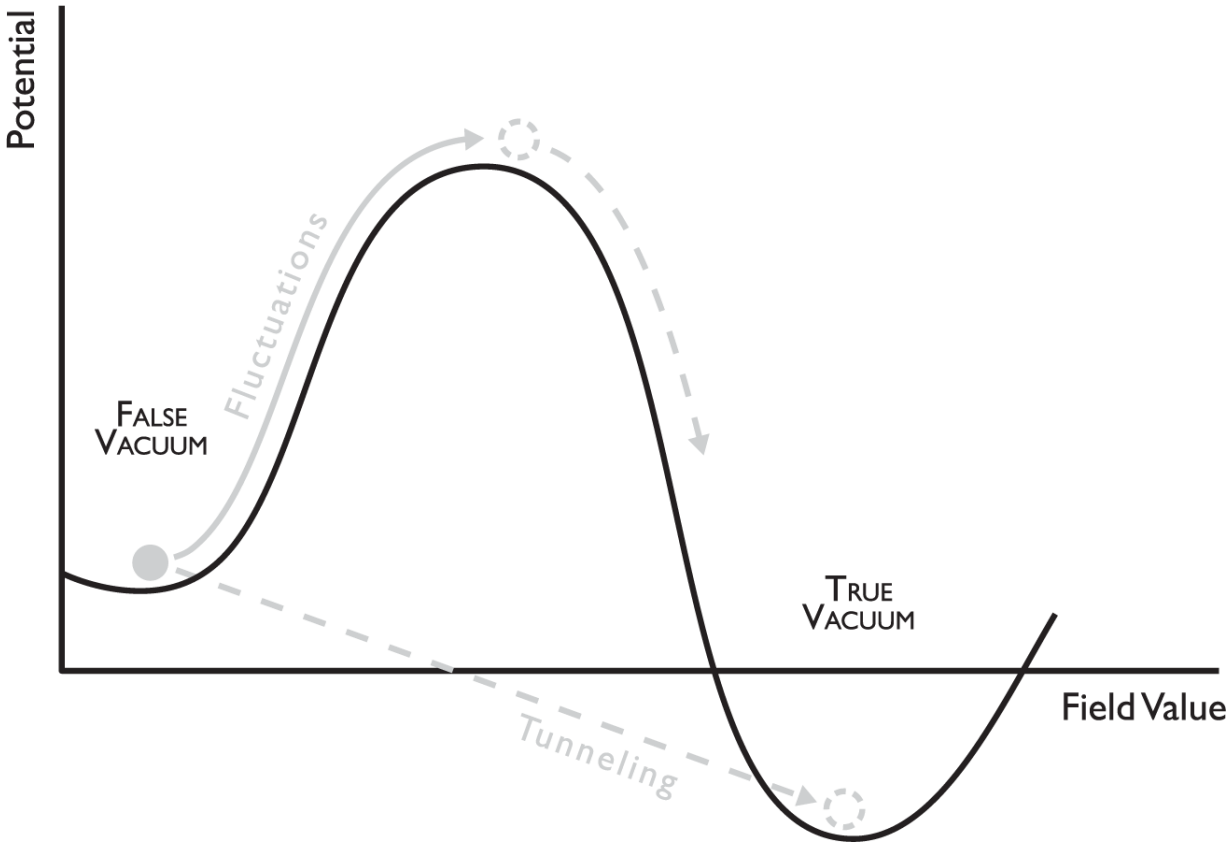


Figure 17: The potential of the Higgs field with a false vacuum state. Each valley in the potential is a possible state of the universe. If our Higgs field lives in the higher valley (the false vacuum), it could transition to the other state (the true vacuum) via a high-energy event (marked "fluctuations" on the diagram) or via quantum tunneling. If we live in a false vacuum universe, a transition of the Higgs field to the true vacuum would be catastrophic.

Unfortunately, the best data we have, consistent with every measurement of the Standard Model of particle physics, suggests that our Higgs field is currently clinging onto just such a divot. This metastable state is also known as a "false vacuum," as opposed to the "true" vacuum at the bottom of the valley floor.